# Deep Generative Models

## Lecture 11

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# Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

## Variance Exploding SDE (NCSN)

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}, \quad \mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

The variance grows since $\sigma(t)$ is a monotonically increasing function.

## Variance Preserving SDE (DDPM)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

The variance is preserved if $\mathbf{x}(0)$ has unit variance.

*Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020*

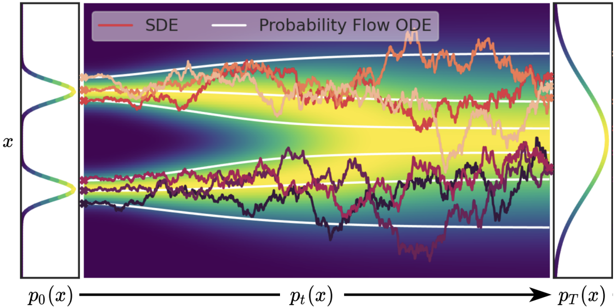
# Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \qquad \text{(SDE with the probability path } p_t(\mathbf{x})\text{)}$$

## Probability Flow ODE

There exists an ODE with the identical probability path $p_t(\mathbf{x})$ of the form:

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \quad \mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$




Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations, 2020*

# Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{v}(\mathbf{x}, t)dt$$

## Reverse ODE

Let $\tau = 1 - t$ $(d\tau = -dt)$:

$$d\mathbf{x} = -\mathbf{v}(\mathbf{x}, 1 - \tau)d\tau$$

## Reverse SDE

There's a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ in the following form:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt + g(t)d\bar{\mathbf{w}}, \quad dt < 0$$

## Sketch of the Proof

- Convert the initial SDE to the probability flow ODE.
- Reverse the probability flow ODE.
- Convert the reverse probability flow ODE to the reverse SDE.

*Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020*

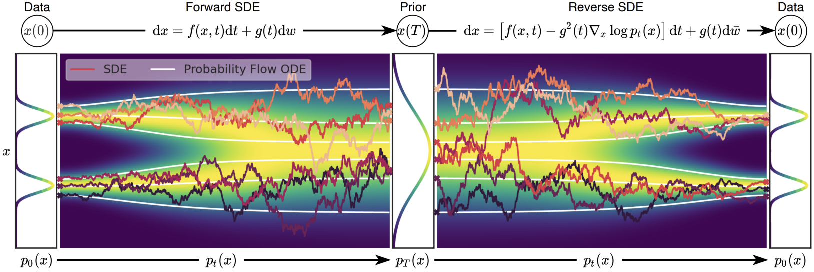
# Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t), \quad \text{where} \quad \mathbf{s}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$$

$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt$ – **Probability Flow ODE**

$d\mathbf{x} = \left(\mathbf{v}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)\right)dt + g(t)d\bar{\mathbf{w}}$ – **Reverse SDE**




*Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020*

# Recap of Previous Lecture

## Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\boldsymbol{\theta}, t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

## Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t) | \mathbf{x}(0))} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t) | \mathbf{x}(0)) \right\|_2^2$$

## NCSN

$$q(\mathbf{x}(t) | \mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0), \left[\sigma^2(t) - \sigma^2(0)\right] \cdot \mathbf{I}\right)$$

## DDPM

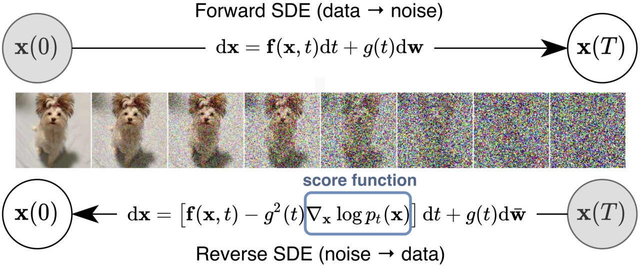
$$q(\mathbf{x}(t) | \mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0) e^{-\frac{1}{2} \int_0^t \beta(s) ds}, \left(1 - e^{-\int_0^t \beta(s) ds}\right) \cdot \mathbf{I}\right)$$

*Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020*

# Recap of Previous Lecture

## Sampling

To sample, solve the reverse SDE using numerical solvers (`SDESolve`).




- Discretizing the reverse SDE gives us ancestral sampling.
- If we use the probability flow ODE instead, then the reverse ODE yields DDIM sampling.

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*Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020*

# Recap of Previous Lecture

Consider ODE dynamics $\mathbf{x}(t)$ in the interval $t \in [0, 1]$ with $\mathbf{x}_0 \sim p_0(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$, $\mathbf{x}_1 \sim p_1(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$.

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

## KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

Solving the continuity equation ODE numerically is complicated and unstable.

## Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

Flow matching is a scalable approach to Neural ODEs.

*Lipman Y., et al. Flow Matching for Generative Modeling, 2022*

# Outline

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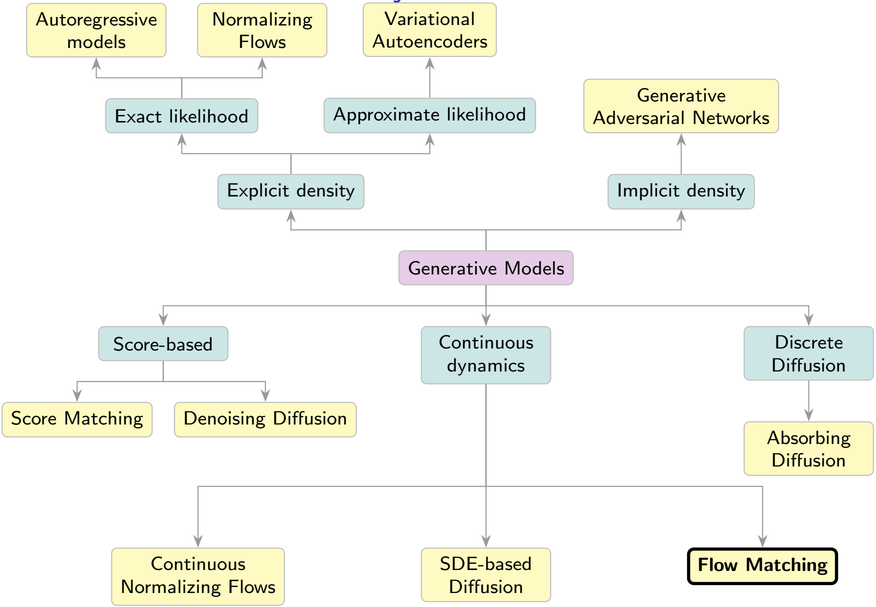
1. Conditional Flow Matching (CFM)
   CFM Objective
   Conditional Probability Paths

2. One-Sided Conditioning

# Generative Models Taxonomy



# Outline

# Flow Matching

Let's introduce the latent variable $\mathbf{z}$:

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Here, $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**.

*Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023*

# Flow Matching

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$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Here, $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**. The conditional probability path $p_t(\mathbf{x}|\mathbf{z})$ satisfies the KFP theorem:

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})\right),$$

where $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**:

*Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023*

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where $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**:

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t) \quad \Rightarrow \quad \frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$$

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# Conditional Vector Field

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) p_t(\mathbf{x}|\mathbf{z})\right),$$

What's the relationship between $\mathbf{v}(\mathbf{x}, t)$ and $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$?

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### Theorem

The following vector field generates the probability path $p_t(\mathbf{x})$:

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \frac{p_t(\mathbf{x}|\mathbf{z}) p(\mathbf{z})}{p_t(\mathbf{x})} d\mathbf{z}$$

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## Proof

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \frac{\partial}{\partial t} \int p_t(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z}$$

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## Proof

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \frac{\partial}{\partial t}\int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z} = \int \left(\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t}\right)p(\mathbf{z})d\mathbf{z}$$

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### Proof

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \frac{\partial}{\partial t} \int p_t(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z} = \int \left( \frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} \right) p(\mathbf{z}) d\mathbf{z} =$$

$$= \int \left(-\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) p_t(\mathbf{x}|\mathbf{z})\right)\right) p(\mathbf{z}) d\mathbf{z}$$

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$$= -\operatorname{div}\left( \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t) p_t(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \right)$$

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## Proof

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \frac{\partial}{\partial t} \int p_t(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z} = \int \left( \frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} \right) p(\mathbf{z}) d\mathbf{z} =$$

$$= \int \left( -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) p_t(\mathbf{x}|\mathbf{z})\right) \right) p(\mathbf{z}) d\mathbf{z} =$$

$$= -\operatorname{div}\left( \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t) p_t(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \right) = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right)$$

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# Flow Matching

## Flow Matching (FM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

## Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

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$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

## Theorem

If $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimal value of the CFM objective.

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# Flow Matching

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$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

## Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

**Theorem**

If $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimal value of the CFM objective.

**Proof**

This can be proved in a similar way as in the denoising score matching theorem.

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# Conditional Flow Matching

## Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]} \mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 =$$

$$= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]} \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2$$

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# Conditional Flow Matching

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$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$

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## Proof

$$\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$

$$= \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left[\|\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\|^2 - 2\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] + \text{const}(\boldsymbol{\theta})$$

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# Conditional Flow Matching

## Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 =$$

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## Proof

$$\mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 =$$

$$= \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left[ \left\| \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 - 2\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) \right] + \text{const}(\boldsymbol{\theta})$$

$$\mathbb{E}_{p_t(\mathbf{x})} \left[ \mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) \right] = \int p_t(\mathbf{x}) \left[ \mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t) \left( \int p_t(\mathbf{z}|\mathbf{x}) \mathbf{v}(\mathbf{x}, \mathbf{z}, t) d\mathbf{z} \right) \right] d\mathbf{x}$$

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# Conditional Flow Matching

## Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$

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## Proof

$$\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$

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$$\mathbb{E}_{p_t(\mathbf{x})} \left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] = \int p_t(\mathbf{x}) \left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t) \left(\int p_t(\mathbf{z}|\mathbf{x})\mathbf{v}(\mathbf{x}, \mathbf{z}, t)d\mathbf{z}\right)\right] d\mathbf{x} =$$

$$= \int\int p_t(\mathbf{x})p_t(\mathbf{z}|\mathbf{x}) \left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, \mathbf{z}, t)\right] d\mathbf{z}d\mathbf{x}$$

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# Conditional Flow Matching

## Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\|^2 =$$

$$= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\|^2$$

## Proof

$$\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\|^2 =$$

$$= \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left[\|\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\|^2 - 2\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] + \mathrm{const}(\boldsymbol{\theta})$$

$$\mathbb{E}_{p_t(\mathbf{x})}\left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] = \int p_t(\mathbf{x}) \left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\left(\int p_t(\mathbf{z}|\mathbf{x})\mathbf{v}(\mathbf{x}, \mathbf{z}, t)d\mathbf{z}\right)\right] d\mathbf{x} =$$

$$= \int\int p_t(\mathbf{x})p_t(\mathbf{z}|\mathbf{x}) \left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, \mathbf{z}, t)\right] d\mathbf{z}d\mathbf{x} =$$

$$= \mathbb{E}_{p(\mathbf{z})}\mathbb{E}_{p_t(\mathbf{x}|\mathbf{z})}\left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, \mathbf{z}, t)\right]$$

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*Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023*

# Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \rightarrow \min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}); \quad \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

*Lipman Y., et al. Flow Matching for Generative Modeling, 2022*

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## Training

1. Sample $t \sim U[0, 1]$, $\mathbf{z} \sim p(\mathbf{z})$, $\mathbf{x}_t \sim p_t(\mathbf{x}|\mathbf{z})$.
2. Compute loss $\mathcal{L} = \left\| \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|^2$.

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## Training

1. Sample $t \sim U[0,1]$, $\mathbf{z} \sim p(\mathbf{z})$, $\mathbf{x}_t \sim p_t(\mathbf{x}|\mathbf{z})$.
2. Compute loss $\mathcal{L} = \left\|\mathbf{v}(\mathbf{x}_t,\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2$.

## Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0,\mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0 = 0, t_1 = 1).$$

*Lipman Y., et al. Flow Matching for Generative Modeling, 2022*

# Marginal-to-Conditional: Scores vs Velocities

## Denoising Score Matching

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Optimal predictor:

$\mathbf{s}_{\boldsymbol{\theta}^*,\sigma}(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)} \left[\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right] = \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$

# Marginal-to-Conditional: Scores vs Velocities

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Optimal predictor: $\mathbf{v}_{\boldsymbol{\theta}^*}(\mathbf{x},t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x},\mathbf{z},t) = \mathbf{v}(\mathbf{x},t)$.

In both cases, the optimal MSE predictor is the posterior mean of the **conditional** target, which equals the intractable **marginal** quantity (score or velocity).

# Outline

# Conditional Flow Matching: Continuity Equations

## Continuity Equations

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) \cdot p_t(\mathbf{x})\right)$$

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) \cdot p_t(\mathbf{x}|\mathbf{z})\right)$$

What is the relationship between $p_t(\mathbf{x})$ and $\mathbf{v}(\mathbf{x}, t)$?

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*Lai C. H. et al. The principles of diffusion models, 2025.*

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- $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.

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- $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.
- But the reverse is not true: any divergence-free vector field $\operatorname{div}(\tilde{\mathbf{v}}(\mathbf{x}, t) p_t(\mathbf{x})) = 0$ can generate a valid probability path.

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- But the reverse is not true: any divergence-free vector field $\operatorname{div}(\tilde{\mathbf{v}}(\mathbf{x}, t)p_t(\mathbf{x})) = 0$ can generate a valid probability path.
- Moreover, not every given path $p_t(\mathbf{x})$ can actually arise from particles moving under some velocity field $\mathbf{v}(\mathbf{x}, t)$.

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# Conditional Flow Matching: Paths and Vector Fields

## Probability Paths

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

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# Conditional Flow Matching: Paths and Vector Fields

## Probability Paths

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

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## Vector fields

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t)\frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})}d\mathbf{z}$$

# Conditional Flow Matching: Paths and Vector Fields

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- $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ and $p_t(\mathbf{x}|\mathbf{z})$ uniquely determine $\mathbf{v}(\mathbf{x}, t)$.
- But marginal velocity does not determine conditional velocities (many decompositions exist).

# From Marginal to Conditional Paths

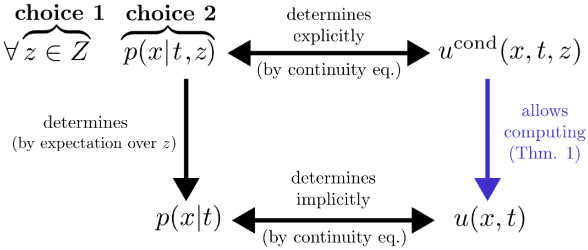


*image credit: A Visual Dive into Conditional Flow Matching*

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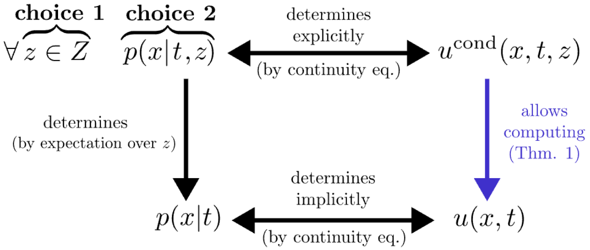


## Open Questions

Q1 How should we choose the convenient conditioning latent variable $\mathbf{z}$?

Q2 How can we parametrize $p_t(\mathbf{x})$ (or $p_t(\mathbf{x}|\mathbf{z})$) to enforce the following constraints?

$$\mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}); \quad \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

*image credit: A Visual Dive into Conditional Flow Matching*

# Gaussian Conditional Probability Paths [Q2]

Let's consider the following parametrization:

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})\right)$$

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The concrete $\boldsymbol{\mu}_t(\mathbf{z})$, $\boldsymbol{\sigma}_t(\mathbf{z})$ (which satisfy the boundary constraints) will be fixed once we choose $\mathbf{z}$ [Q1].

## Gaussian flow

- Let's derive the vector field for **any** Gaussian path $p_t(\mathbf{x}|\mathbf{z})$.
- There are infinitely many flows $\boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z})$ (or, equivalently, vector fields $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$) that generate a particular probability path $p_t(\mathbf{x}|\mathbf{z})$.

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- Let's consider the following flow:

$$\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z}) = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon},$$

where $\boldsymbol{\epsilon}$ is fixed by the condition: $\mathbf{x}_0 = \boldsymbol{\mu}_0(\mathbf{z}) + \boldsymbol{\sigma}_0(\mathbf{z}) \odot \boldsymbol{\epsilon}$.

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# Gaussian Conditional Vector Field

## Gaussian Conditional Probability Path

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How to derive the expression for $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$ for this particular flow $\boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z})$?

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$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t); \quad \boldsymbol{\epsilon} = \frac{1}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))$$

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$$\boxed{\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) = \boldsymbol{\psi}_t'\left(\boldsymbol{\psi}_t(\mathbf{x}_t, \mathbf{z})^{-1}, \mathbf{z}\right) = \boldsymbol{\mu}_t'(\mathbf{z}) + \frac{\boldsymbol{\sigma}_t'(\mathbf{z})}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))}$$

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# Outline

# Endpoint Conditioning

### Conditional Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \rightarrow \min_{\boldsymbol{\theta}}$$

Let's define our latent variable $\mathbf{z}$.

*Lipman Y., et al. Flow Matching for Generative Modeling, 2022*

# Endpoint Conditioning

## Conditional Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

Let's define our latent variable $\mathbf{z}$.

## Conditioning Latent Variable [Q1]

Let us choose $\mathbf{z} = \mathbf{x}_1$. Then $p(\mathbf{z}) = p_1(\mathbf{x}_1)$.

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{x}_1) p_1(\mathbf{x}_1) d\mathbf{x}_1$$

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# Endpoint Conditioning

## Conditional Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

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We need to ensure the boundary constraints:

$$\begin{cases} \mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}) \\ \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}) \end{cases}$$

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*Lipman Y., et al. Flow Matching for Generative Modeling, 2022*

# One-Sided Conditioning

$$p_0(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(0, \mathbf{I}); \quad p_1(\mathbf{x}|\mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

## Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_1)\right); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

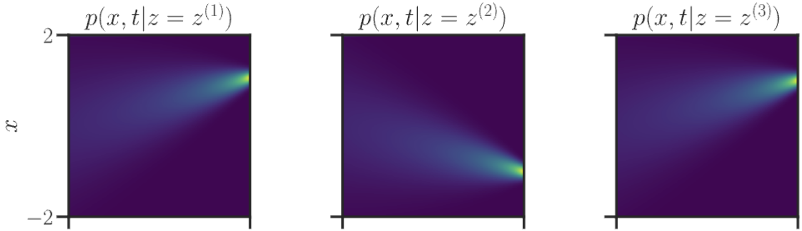
*image credit: A Visual Dive into Conditional Flow Matching*

# One-Sided Conditioning

$$p_0(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(0, \mathbf{I}); \quad p_1(\mathbf{x}|\mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

## Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_1)\right); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

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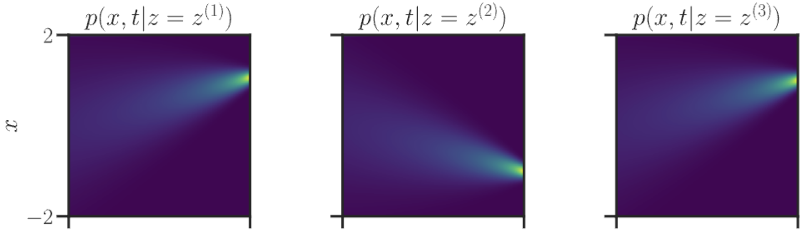


Let's consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 \\ \boldsymbol{\sigma}_t(\mathbf{x}_1) = 1 - t \end{cases}$$

---

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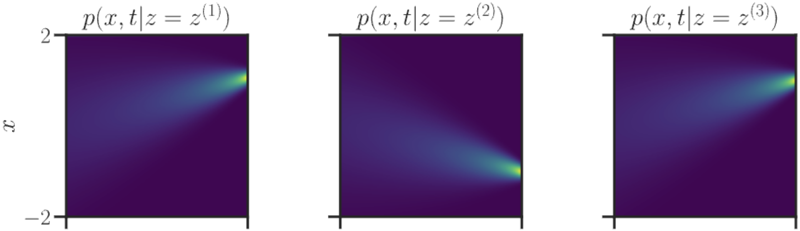


Let's consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 \\ \boldsymbol{\sigma}_t(\mathbf{x}_1) = 1 - t \end{cases} \Rightarrow \begin{cases} p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2 \cdot \mathbf{I}\right) \\ \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0 \end{cases}$$

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# One-Sided Conditioning: Conditional Vector Field

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2\mathbf{I}\right); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

## Conditional Vector Field

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \boldsymbol{\mu}'_t(\mathbf{x}_1) + \frac{\boldsymbol{\sigma}'_t(\mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_1))$$

*image credit: https://mlg.eng.cam.ac.uk/blog/2024/01/20/flow-matching.html*

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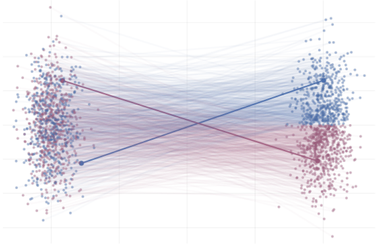

The conditional vector field $\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t)$ defines straight lines between $p_{\text{data}}(\mathbf{x})$ and $\mathcal{N}(0, \mathbf{I})$.

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*image credit: https://mlg.eng.cam.ac.uk/blog/2024/01/20/flow-matching.html*

# One-Sided Conditioning: CFM Objective

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2 =$$

$$= \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{x}_1)}\left\|\frac{\mathbf{x}_1-\mathbf{x}}{1-t}-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

Lipman Y., et al. *Flow Matching for Generative Modeling, 2022*

# One-Sided Conditioning: CFM Objective

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 =$$

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$$= \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})} \left\| (\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\boldsymbol{\theta}} \left( t\mathbf{x}_1 + (1 - t)\mathbf{x}_0, t \right) \right\|^2$$

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- We fit straight lines between the noise distribution $p_0(\mathbf{x})$ and the data distribution $p_{\text{data}}(\mathbf{x})$.
- The **marginal** path $p_t(\mathbf{x})$ does not give straight lines.

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# CFM with One-Sided Conditioning: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(0,\mathbf{I})} \left\| (\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

## Training

1. Sample $\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})$, $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$, $t \sim U[0, 1]$.
2. Compute noisy image $\mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$.
3. Compute loss $\mathcal{L} = \left\| (\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|^2$.

*Lipman Y., et al. Flow Matching for Generative Modeling, 2022*

# CFM with One-Sided Conditioning: Training and Sampling

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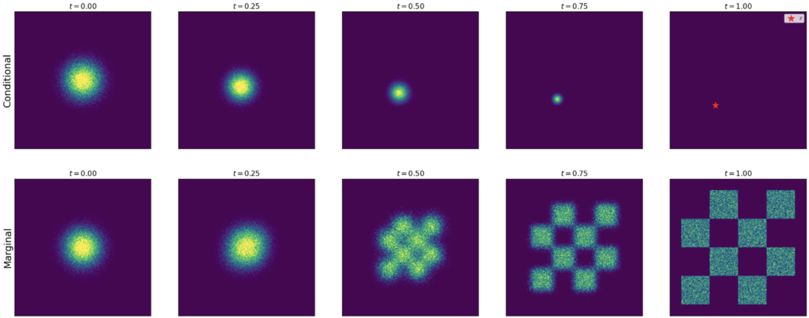
## Sampling

1. Sample $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1=\psi_1(\mathbf{x}_0)=\texttt{ODESolve}_v(\mathbf{x}_0,\boldsymbol{\theta},t_0=0,t_1=1).$$

*Lipman Y., et al. Flow Matching for Generative Modeling, 2022*

# Conditional vs. Marginal Paths

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*Holderrieth P. and Erives E., et al. An Introduction to Flow Matching and Diffusion Models, 2025*

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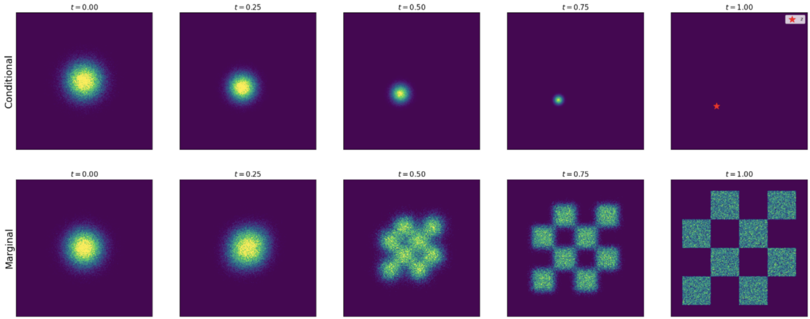


- One-sided conditioning gives us the way to construct a generative model.
- Now we extend it to image-to-image formulation (mapping between two distinct data distributions $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$).

*Holderrieth P. and Erives E., et al. An Introduction to Flow Matching and Diffusion Models, 2025*

# Summary

- Conditional flow matching introduces the latent variable $\mathbf{z}$, reformulating the initial task in terms of conditional dynamics.

- The marginal vector field can be expressed as the posterior mean of conditional vector fields: $\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$.

- The FM and CFM objectives have the same optimal solution, making CFM a tractable alternative to FM.

- For Gaussian conditional paths $p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z}))$, the conditional vector field has a closed form for $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$.

- One-sided conditioning uses endpoint conditioning $\mathbf{z} = \mathbf{x}_1$ with straight conditional paths, yielding the simple vector field $\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \mathbf{x}_1 - \mathbf{x}_0$.